

ADAMTS13, Recombinant-krhn (Adzynma) - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses ADAMTS13, recombinant-krhn (Adzynma) for commercial medical plans. For Medicare criteria, see [Medicare Part B Criteria](#).

Note: Requires Precertification:

Precertification of ADAMTS13, recombinant-krhn (Adzynma) is required of all Aetna participating providers and members in applicable plan designs. For precertification of ADAMTS13, recombinant-krhn (Adzynma), call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see [Specialty Pharmacy Precertification](#).

Note: Site of Care Utilization Management Policy applies. For information on site of service for ADAMTS13, recombinant-krhn (Adzynma), see [Utilization Management Policy on Site of Care for Specialty Drug Infusions](#).

1. Criteria for Initial Approval

Congenital thrombotic thrombocytopenic purpura (cTTP)

Aetna considers ADAMTS13, recombinant-krhn (Adzynma) medically necessary for the treatment of congenital thrombotic thrombocytopenic purpura (cTTP) when *both* of the following criteria are met:

1. The diagnosis of cTTP has been confirmed by genetic testing with biallelic mutations in the ADAMTS13 gene; *and*
2. Member has an ADAMTS13 activity level of less than 10% at the time of diagnosis.

Aetna considers all other indications as experimental, investigational, or unproven.

2. Continuation of Therapy

Aetna considers continuation of ADAMTS13, recombinant-krhn (Adzynma) therapy medically necessary in members requesting reauthorization for an indication listed in Section I who are responding to therapy (e.g., reduction or maintenance of number of thrombotic thrombocytopenic purpura [TTP] events, increase in platelet count, decrease in lactate

dehydrogenase [LDH] level).

Dosage and Administration

ADAMTS13, recombinant-krhn is available as Adzynma as a lyophilized powder in single-dose vials containing nominally 500 or 1500 international units for intravenous use after reconstitution only.

The recommended dosage and administration for Adzynma is as follows:

- Prophylactic therapy
- Administer 40 IU/kg body weight once every other week intravenously at a rate of 2 to 4 mL per minute.
- The prophylaxis dosing frequency may be adjusted to 40 IU/kg body weight once weekly based on prior prophylactic dosing regimen or clinical response.
- On-demand therapy - Administer intravenously at a rate of 2 to 4 mL per minute:
 - 40 IU/kg body weight on day 1.
 - 20 IU/kg body weight on day 2.
 - 15 IU/kg body weight on day 3 and beyond until two days after the acute event is resolved.

Refer to the full prescribing information for Adzynma for preparation and administration instructions.

Source: Takeda Pharmaceuticals U.S.A., 2023

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
85397	Coagulation and fibrinolysis, functional activity, not otherwise specified (eg, ADAMTS-13), each analyte
Other CPT codes related to the CPB:	
83615	Lactate dehydrogenase (LD), (LDH)
83625	Lactate dehydrogenase (LD), (LDH); isoenzymes, separation and quantitation
85049	Blood count; platelet, automated
96374	Therapeutic, prophylactic, or diagnostic injection (specify substance or drug); intravenous push, single or initial substance/drug
HCPCS codes covered if selection criteria are met:	
J7171	Injection, adamts13, recombinant-krhn, 10 iu
ICD-10 codes covered if selection criteria are met:	
D69.42	Congenital and hereditary thrombocytopenia purpura

Background

U.S. Food and Drug Administration (FDA)-Approved Indications

- Adzynma is indicated for prophylactic or on demand enzyme replacement therapy (ERT) in adult and pediatric patients with congenital thrombotic thrombocytopenic purpura (cTTP).

ADAMTS13, recombinant-krhn is available as Adzynma (Takeda Pharmaceuticals U.S.A., Inc.). Adzynma is a purified bivalent human recombinant "A disintegrin and metalloproteinase with thrombospondin motifs 13" (rADAMTS13) expressed in Chinese Hamster Ovary (CHO) cells using recombinant DNA technology (a mixture of Native rADAMTS13 Q23 and Variant rADAMTS13 R23 with a controlled range of the two variants ratio). Adzynma is a recombinant form of the endogenous ADAMTS13. ADAMTS13 is a plasma zinc metalloprotease that controls the activity of von Willebrand factor (VWF) by cleaving large and ultra-large VWF multimers to smaller units and thereby reducing the platelet binding properties of VWF and its tendency to form microthrombi (Takeda Pharmaceuticals U.S.A., 2023).

According to the prescribing information, Adzynma use is contraindicated in patients who have manifested life threatening hypersensitivity reactions to Adzynma or its components.

Adzynma carries the following warnings and precautions per the prescribing information:

- Hypersensitivity reactions may occur. Discontinue Adzynma if hypersensitivity symptoms occur and administer appropriate emergency treatment.
- Immunogenicity: Patients may develop antibodies to rADAMTS13 which could potentially result in a decreased or lack of response to rADAMTS13. Patients may develop antibodies to host cell proteins which could potentially result in adverse reactions. There are no data on risk in previously untreated patients (subjects naive to plasma-based products).

Adzynma carries the following most common adverse reactions (incidence > 5%) per the prescribing information: headache, diarrhea, migraine, abdominal pain, nausea, upper respiratory tract infection, dizziness, and vomiting.

Congenital thrombotic thrombocytopenic purpura (cTTP) is a very rare, inherited blood clotting disorder characterized by blood clot formation in the small blood vessels throughout the body. This blood disorder is caused by a disease-causing mutation in the ADAMTS13 gene, which is tasked with making an enzyme, ADAMTS13, that regulates blood clotting. The ADAMTS13 gene mutation follows an autosomal recessive inheritance pattern. In the U.S., cTTP affects an estimate of fewer than one thousand people. Signs and symptoms typically develop in infancy or early childhood, but in some cases may develop in adulthood and may first become apparent during pregnancy. Individual with cTTP may experience severe bleeding episodes, strokes, and damage to vital organs. The disease can be fatal if left untreated. Treatment for cTTP usually involves prophylactic plasma-based therapy for individuals with chronic disease to decrease the risk of clotting/bleeding by supplying the absent/low ADAMTS13 enzyme (FDA, 2023; NORD, 2024).

On November 9, 2023, U.S. Food and Drug Administration (FDA) approved Adzynma (ADAMTS13, recombinant-krhn) for prophylactic and on-demand enzyme replacement therapy (ERT) of adult and pediatric patients with congenital thrombotic thrombocytopenic purpura (cTTP). Adzynma is the first FDA-approved recombinant ADAMTS13 (rADAMTS13) protein designed for patients with cTTP by replacing the deficient ADAMTS13 enzyme. The FDA approval was based on supporting data from a randomized, controlled, open-label, crossover phase 3 trial evaluating the safety and effectiveness of prophylactic and on-demand ERT with Adzynma compared to plasma-based therapies in patients with cTTP. The efficacy of Adzynma in the prophylactic treatment of patients with cTTP was evaluated in 46 patients who were randomized to receive 6 months of treatment with either 40 IU/kg of Adzynma or plasma-based therapies (Period 1), then crossed over to the other treatment for 6 months (Period 2). Thirty-five patients entered the 6-month single arm period with Adzynma (Period 3). No patients receiving Adzynma had an acute TTP event throughout the study, including Period 3. No subacute TTP events were reported in patients receiving Adzynma during Periods 1 and 2. The mean annualized event rate of thrombocytopenia manifestations was 2.0 for patients receiving Adzynma compared to 4.44 in patients receiving plasma-based therapies (FDA, 2023; Takeda Pharmaceuticals U.S.A., 2023a). In a pharmacokinetic assessment, patients receiving Adzynma achieved a four-to five-fold increase in ADAMTS13 activity following a single infusion compared to plasma-based therapies (Takeda Pharmaceuticals U.S.A., 2023b).

References

The above policy is based on the following references:

1. Asmis LM, Serra A, Krafft A, et al. Recombinant ADAMTS13 for hereditary thrombotic thrombocytopenic purpura. *N Engl J Med*. 2022;387(25):2356-2361.
2. National Organization for Rare Disorders (NORD). Congenital thrombotic thrombocytopenic purpura. Rare Disease Database. Quincy, MA: NORD; 2024. Available at: <https://rarediseases.org/gard-rare-disease/congenital-thrombotic-thrombocytopenic-purpura/>. Accessed January 4, 2024.
3. Scully M, Antun A, Ctalan SR, et al. Recombinant ADAMTS13 in congenital thrombotic thrombocytopenic purpura. *N Engl J Med* 2024; 390(17):1584-1596.
4. Takeda Pharmaceuticals U.S.A., Inc. Adzynma (ADAMTS13, recombinant-krhn) lyophilized powder for injection, for intravenous use. Prescribing Information. Lexington, MA: Takeda Pharmaceuticals U.S.A.; revised November 2023a.
5. Takeda Pharmaceuticals U.S.A., Inc. Adzynma (ADAMTS13, recombinant-krhn) lyophilized powder for injection, for intravenous use. Prescribing Information. Lexington, MA: Takeda Pharmaceuticals U.S.A.; revised June 2024.
6. Takeda Pharmaceuticals U.S.A., Inc. Takeda's Adzynma (ADAMTS13, recombinant-krhn) approved by U.S. FDA as the first and only recombinant ADAMTS13 enzyme replacement therapy for the treatment of congenital thrombotic thrombocytopenic purpura (cTTP). Press Release. Lexington, MA: Takeda Pharmaceuticals U.S.A.; November 9, 2023b.
7. U.S. Food and Drug Administration (FDA). FDA approves first treatment for patients with rare inherited blood clotting disorder. FDA News Release. Silver Spring, MD: FDA; November 9, 2023.